

Karthigeyan Ganesh Shankar

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WORK EXPERIENCE

Senior Solutions Engineer

June 2026 – Present

Vimaan AI, India ([Website](#))

Bengaluru, India

- Coordinated as a **primary liaison** between operations and engineering, implementing best practices that reduced engineering workload (30%) while maintaining project delivery timelines
- Engineered a cross-functional partner integration with key stakeholders, delivering a **production-ready solution** ahead of schedule without sacrificing reliability or system robustness
- Perfected **firmware, software, geometric and computer vision pipelines** for three different product SKUs at Vimaan to conform to customer use-cases by bringing in engineering acumen
- Improved tooling and awareness with **powerful token reduction methods (67%)** of Claude and advocated mindful usage of AI tools for engineering workflows. Mentored engineers to imbibe them
- **Tool Stack:** [Foxglove](#), [ROS1](#), [Neo4j](#), [Kafka](#) [Triton Inference Server](#)

Lead Autonomy Engineer

April 2025 – May 2026

Ati Motors, India ([Website](#))([Blog](#))

Bengaluru, India

- Pioneered organization wide use (Solutions, Validation and Dev teams) of optimized **simulation-first** development and testing. Made the workflow available for Cloud, on-prem and edge environments
- Inspired the map-free navigation models which are generalized and SOTA (**NoMad, ViNT**), proactively identifying challenges before porting it to production and tweaking for Ati's vehicle archetypes
- Investigated the use of novel, SOTA 3D reconstruction (**VGGT, Mast3R, Splatfacto**) and VSLAM methods (**Mast3R-SLAM, Droid-SLAM**) to migrate towards vision based navigation and representation
- **Tool Stack:** [Nvidia Isaac Sim](#), [Nvidia Warp](#), [Pytorch](#), [Blender](#), [MeshLab](#), [Cloudcompare](#)

Senior Autonomy Engineer

April 2024 – April 2025

Ati Motors, India ([Website](#))

Bengaluru, India

- Spearheaded a team which developed an one button press simulation workflow by aligning the current software stack with **Nvidia's Isaac Sim** application. The effort entails setting up **sim2real** environment and emulating existing control, navigation and perception stack of the bots into simulation
- Solely responsible for evaluating different OTS carrier boards (**Jetson**) that controls the entire bot by semi-automating the bring-up process and saving considerable dev and validation time
- Built business critical, **production ready architecture** and implementation framework for both the bot and fleet manager application and making factual recommendations
- **Tool Stack:** [Nvidia Isaac Sim](#), [Kubernetes](#), [Bazel](#), [Python](#), [Make](#), [Unix](#), [Embedded C](#)

Autonomy Engineer

Sept 2022 – March 2024

Ati Motors, India ([Website](#))

Bengaluru, India

- Revived and optimized the software build systems to catalyse the build process by **40 %** enabling the team to deploy features/bug fixes at speed with utmost reliability
- Developed the software-in-loop (CI/CD) frameworks to **automate** the software release process and reduce process bottlenecks with **bash scripting** and offline testing exigencies
- Enabled the team to use designed systems across all available software architectures increasing their productivity and throughput by **30%** by provided necessary training to use them well
- Supported in code development, robotic **deployment** and gained a working understanding of the industrial market and product development cycle of different products as part of the portfolio
- **Tool Stack:** [Kubernetes](#), [Bazel](#), [Poetry](#), [Docker](#), [C++](#), [Python](#), [Make](#), [Unix](#), [Bash](#)

Research Assistant

Jul 2019 – Mar 2020

Institute for Geometric and Applied Mathematics, RWTH Aachen ([Website](#))

Aachen, Germany

- Implemented data fitting models for **250,000+ data points** for **six** different distributions and determined the best fit. The pattern emerged as biased pareto distributions
- **Software/Language Used:** [Matlab](#), [Python](#), [Seaborn](#), [Numpy](#)

Research Assistant

Institute for Aerodynamics and Fluid Mechanics, RWTH Aachen ([Website](#))

Oct 2018 – Mar 2019
Aachen, Germany

- Devised massively parallel simulations in C++ for **1 million cells through 24 cores** for **100+ core hours** to determine the optimal configuration for a laminar pre-mixed flame propagation.
- **Tool Stack:** [C++](#), [Python](#), [ParaView](#), [Unix](#), [Bash](#), [OpenMPI](#)

Head of Growth (April 2022); Customer Success Engineer

Gradly ([Website](#))

Jun 2021 – Jul 2022
Bengaluru, India

- Achieved **40% activation/retention** of customers and designed automation systems and boosted the **overall efficiency by 30%**
- Acquired **200+ customers** and processed **100+ support tickets**
- Co-developed the insurance backend and knowledge base to efficiently service customers and yielded a net revenue of **\$100k+ in sales**
- **Tool Stack:** [Notion](#), [Figma](#), [Linear](#), [Zapier](#), [ParseHub](#), [Airtable](#), [Canva](#), [Front](#)

Research Intern

Heat Transfer Lab, Indian Institute of Technology Guwahati ([Website](#))

June 2016 – July 2016
Guwahati, India

Intern

Devi Polymers Pvt. Ltd. ([Website](#))

June 2015 – July 2015
Chennai, India

EDUCATION

Rheinisch-Westfälische Technische Hochschule (RWTH) Aachen

Master of Sciences in Simulation Sciences

Oct 2017 – Mar 2020
Aachen, Germany

SRM University

Bachelor of Technology in Mechanical Engineering

May 2013 – May 2017
Chennai, India

SKILLS

Programming Languages: Python, C++, MATLAB, Bash, Rust (Beginner)

Tools: Dofin, ScoreP, $\LaTeX$, Gmsh, ParaView, NumPy, SciPy, Docker, Poetry, Blender, Nvidia Isaac Sim, MeshLab, CloudCompare, Bazel, Makefile, Kubernetes, Warp

Languages: English (Fluent), French (Intermediate), German (Beginner), Hindi (Intermediate), Tamil (Native)

OTHER EXPERIENCES

Repository homepage: [GitHub](#)

Master Thesis: Kinetic Model on Unemployment, RWTH Aachen

- Conceptualized and developed **mesoscopic models** to spatially and temporally characterize unemployment
- Simulated and characterized the interactive, time-variant complex system and analytically concluded the upper bounds

Bachelor Thesis: Fatigue Characteristics of Functionally Graded Materials, SRM University

- Supervised manufacturing of **Aluminium (L24)** with different filler materials (titanium oxide, silicon carbide, aluminium oxide) at **under-engineered locations** based on their stress profiles
- Administered and performed experimental analysis to obtain **natural frequency** of the manufactured material using a tri-axial accelerometer and subsequently fatigue frequency was calculated with respect to the frequency of Aluminium